A Smoothness Covering Decomposition:
Covering $[1, n]$ by Divisibility with One Dedicated Element per Large Prime Power

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Abstract

We give an explicit, elementary construction of a set D of positive integers that covers $[1, n]$ by divisibility — every $i \le n$ divides some element of D — using a single "smooth" element together with one dedicated element per prime power in $(\sqrt{n}, n]$. The construction has size $|D| = \pi(n)\,(1 + o(1))$ and its elements are of magnitude $\exp\big(O(\sqrt{n})\big)$. The proof rests on a single observation: no integer $i \le n$ has two coprime maximal prime-power divisors both exceeding $\sqrt{n}$. The result isolates the prime powers above $\sqrt{n}$ as the sole carriers of covering cost: every $\sqrt{n}$-smooth number and every integer with a small cofactor is absorbed for free by one shared element.

1 Setup

Definition 1. For a finite set $D \subset \mathbb{Z}_{>0}$, we say D *covers* $[1, n]$ *by divisibility* if for every integer i with $1 \le i \le n$ there exists $d \in D$ with $i \mid d$.

Throughout, $n \ge 2$ is an integer, $t = \lfloor \sqrt{n} \rfloor$, and

$$L \;=\; \mathrm{lcm}(1, 2, \ldots, t).$$

For an integer $i \ge 1$ we call p^a a *maximal prime power of* i if p is prime, $a \ge 1$, and $p^a \parallel i$ (that is, $p^a \mid i$ but $p^{a+1} \nmid i$). The maximal prime powers of i are pairwise coprime and their product equals i.

2 The decomposition

Theorem 1 (Smoothness Covering Decomposition)**.** *Let $n \ge 2$, $t = \lfloor \sqrt{n} \rfloor$, and $L = \mathrm{lcm}(1, \ldots, t)$. Let*

$$Q \;=\; \{\, q = p^a \;:\; p \text{ prime},\; a \ge 1,\; \sqrt{n} < q \le n \,\}$$

be the set of prime powers in $(\sqrt{n}, n]$. Then

$$D \;=\; \{L\} \,\cup\, \{\, L \cdot q \;:\; q \in Q \,\}$$

covers $[1, n]$ by divisibility, and

$$|D| \;=\; 1 + |Q| \;=\; \pi(n)\,\big(1 + o(1)\big).$$

The proof uses the following elementary lemma.

Lemma 1. *Every integer i with $1 \le i \le n$ has at most one maximal prime power exceeding $\sqrt{n}$.*

Proof. Suppose $q_1 = p_1^{a_1}$ and $q_2 = p_2^{a_2}$ are two distinct maximal prime powers of i, both greater than $\sqrt{n}$. Distinct maximal prime powers are coprime, so $q_1 q_2 \mid i$, whence $i \ge q_1 q_2 > \sqrt{n} \cdot \sqrt{n} = n$, contradicting $i \le n$. $\square$

Proof of Theorem 1. Fix i with $1 \le i \le n$. By Lemma 1, either all maximal prime powers of i are at most $\sqrt{n}$, or exactly one exceeds $\sqrt{n}$.

Case 1: every maximal prime power of i is $\le \sqrt{n}$. Each maximal prime power $p^a \parallel i$ satisfies $p^a \le \sqrt{n} \le t$, hence $p^a \mid L$ because $L = \operatorname{lcm}(1, \dots, t)$ is divisible by every integer up to t. The maximal prime powers of i are pairwise coprime and their product is i, so their product divides L; that is, $i \mid L$. Thus i divides the element $L \in D$.

Case 2: exactly one maximal prime power $q = p^a$ of i satisfies $q > \sqrt{n}$. Then $q \in Q$ (as $q \mid i \le n$ gives $q \le n$), and $L \cdot q \in D$. Write $i = q\,m$ with $m = i/q$. Since $q > \sqrt{n}$,

$$m = \frac{i}{q} \le \frac{n}{q} < \frac{n}{\sqrt{n}} = \sqrt{n} \le t,$$

so $m \le t$ and therefore $m \mid L$. Moreover $q = p^a$ is the *full* power of p dividing i, so $p \nmid m$, giving $\gcd(m, q) = 1$. From $m \mid L$ and $q \mid q$ with $\gcd(m, q) = 1$ we obtain $m\,q \mid L\,q$. Since $i = m\,q$, it follows that $i \mid L \cdot q$.

In either case i divides an element of D, so D covers $[1, n]$.

For the count, $|D| = 1 + |Q|$ by construction. The elements of Q are the prime powers in $(\sqrt{n}, n]$. Prime powers p^a with $a \ge 2$ and $p^a \le n$ number $O(\sqrt{n})$ in total, while the primes in $(\sqrt{n}, n]$ number $\pi(n) - \pi(\sqrt{n}) = \pi(n)(1 + o(1))$. Hence $|Q| = \pi(n)(1 + o(1))$ and $|D| = \pi(n)(1 + o(1))$. $\square$

3 Remarks

Remark 1 (Magnitude). Since $\log L = \psi(t) = \psi(\lfloor \sqrt{n} \rfloor) \sim \sqrt{n}$ by the prime number theorem (where ψ is the second Chebyshev function), every element of D has magnitude

$$\max_{d \in D} d \;=\; L \cdot \max Q \;\le\; L \cdot n \;=\; \exp\bigl(\sqrt{n}\,(1 + o(1))\bigr).$$

Remark 2 (What the decomposition isolates). The single element L absorbs, for free, every $\sqrt{n}$-smooth integer and, via Case 2, every integer of the form $m\,q$ with $m \le \sqrt{n}$ and q a prime power above $\sqrt{n}$. Consequently the only integers requiring their own dedicated element are handled through the $\sim \pi(n)$ prime powers in $(\sqrt{n}, n]$. In this sense the prime powers above $\sqrt{n}$ are the sole carriers of covering cost in the construction.

Remark 3 (Verification). The case analysis in the proof was checked exhaustively for every $i \le n$ and all n with $2 \le n \le 3000$: in each instance the predicted covering element (L in Case 1, $L \cdot q$ in Case 2) was confirmed to be divisible by i, the “at most one large maximal prime power” lemma held, and the full set D was confirmed to cover $[1, n]$.